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A double auxiliary optimization constrained multi-objective evolutionary algorithm

Yongkuan Yang^{*}, Bing Yan, Xiangsong Kong, Jing Zhao

School of Electrical Engineering and Automation, Xiamen University of Technology, Xiamen 361024, China
 Xiamen Key Laboratory of Frontier Electric Power Equipment and Intelligent Control, Xiamen 361024, China

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ABSTRACT

In evolutionary constrained multi-objective optimization, the use of auxiliary optimization is gradually attracting attention. It is noted that different forms of auxiliary optimization have different advantages. Combining these advantages in an appropriate manner can further improve the algorithm's performance. Motivated by this inspiration, we propose a double auxiliary optimization constrained multi-objective evolutionary algorithm, namely DAO. In DAO, two auxiliary optimizations, i.e., the unconstrained optimization and the (M+1)-objective optimization, are applied in a proposed tri-population co-evolution strategy. In this strategy, three populations are used to optimize the core optimization and the two auxiliary optimizations in an interactive evolution form. Furthermore, DAO develops a (M+1)-objective environmental selection strategy to deeply explore the boundary between the feasible and infeasible regions. In experimental studies, the performance of DAO and four other state-of-the-art algorithms is evaluated on three distinct benchmark test suites and two intricate real-world problem scenarios. The outcome of the comprehensive evaluation shows the competitive of DAO solving constrained multi-objective optimization problems.

1. Introduction

Many practical optimization problems, such as aerodynamic shape optimization [25], web service location allocation [33], and gas pipeline operation [6] can be seen as constrained multi-objective optimization problems (CMOPs) [16], which can be mathematically represented by the following equations.

$$\begin{aligned} \min \quad & \mathbf{F}(\mathbf{x}) = \{f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x})\}, \\ \text{s.t.} \quad & g_j(\mathbf{x}) \leq 0, j = 1, \dots, J, \\ & h_k(\mathbf{x}) = 0, k = 1, \dots, K, \end{aligned} \quad (1)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_N) \in \mathbb{R}^N$ is a decision vector (solution) including N decision variables; $\mathbf{F}(\mathbf{x}) \in \mathbb{R}^M$ is a objective vector including M objective functions; $g_j(\mathbf{x})$ means the number j inequality constraint; $h_k(\mathbf{x})$ means the number k equality constraint; K and J are the quantities of equality and inequality constraints, respectively. The extent of constraint violation $\phi(\mathbf{x})$ exhibited by a solution $\mathbf{x}$ can be precisely characterized through the following equation.

$$\phi(\mathbf{x}) = \sum_{j=1}^J \max\{0, g_j(\mathbf{x})\} + \sum_{k=1}^K \max\{0, |h_k(\mathbf{x}) - \delta|\}, \quad (2)$$

^{*} Corresponding author at: School of Electrical Engineering and Automation, Xiamen University of Technology, Xiamen 361024, China.
 E-mail address: yangyongkuanneu@sina.com (Y. Yang).

where δ is a small positive threshold (e.g., $\delta=10^{-4}$). When $\phi(\mathbf{x})$ exceeds zero ($\phi(\mathbf{x}) > 0$), it categorizes $\mathbf{x}$ as an infeasible solution. On the other hand, if $\phi(\mathbf{x})$ equals zero ($\phi(\mathbf{x}) = 0$), $\mathbf{x}$ is considered a feasible solution.

We say that a solution $\mathbf{x} \in \mathbb{R}^N$ dominates another solution $\mathbf{y} \in \mathbb{R}^N$ (denoted by $\mathbf{x} < \mathbf{y}$) if $f_i(\mathbf{x}) < f_i(\mathbf{y})$ in at least one $i \in \{1, \dots, M\}$ and $f_j(\mathbf{x}) \leq f_j(\mathbf{y})$ for each $j \in \{1, \dots, M\}$. The characteristic of a Pareto optimal solution, denoted as $\mathbf{x}^*$ is that it remains undominated by any other solution within the given objective space. All the Pareto optimal solutions form the Pareto set. The Pareto front (PF) is the image of the Pareto set in the objective space [19,34]. It is noted that Pareto optimal solutions for CMOPs are the Pareto optimal feasible solutions. [15,23,24,29].

Evolutionary algorithms (EAs) are widely utilized for solving complex optimization problems due to their parallel search capability inspired by nature. Numerous constrained multi-objective EAs (CMOEAs) have been proposed to tackle CMOPs [1,3,7,47,48]. Since the optimal solution for a CMOP is a feasible solution set located on the constrained Pareto front (PF), the core of CMOEAs lies in effectively balancing the convergence, distribution, and feasibility of solutions through constraint handling techniques (CHTs) [13,39]. Based on the number of populations used in CHTs, current CMOEAs can be categorized into single-population CMOEAs and multi-population CMOEAs. In multi-population CMOEAs, one approach involves solving the CMOPs with the assistance of an auxiliary optimization problem. Consequently, two populations evolve together to simultaneously solve the original CMOP and the auxiliary optimization problem. There is no doubt that the formulation of the auxiliary optimization problem and the approach to collaborative evolution significantly impact the algorithm's performance. And different forms of auxiliary optimization have different advantages. Combining these advantages in an appropriate manner can further improve the algorithm's performance. Motivated by this inspiration, this paper proposes a dual auxiliary optimization CMOEA. The main contributions of this article can be summarized as follows:

1. To facilitate the solving of the original CMOP, two auxiliary optimization tasks are concurrently established: the $(M+1)$ -objective optimization task and the unconstrained optimization task.
2. A tri-population co-evolution strategy is developed to optimize the three optimization problems.
3. A $(M+1)$ -objective environmental selection strategy is proposed to deeply explore the boundary between feasible and infeasible regions.

The subsequent sections of this paper are structured as follows: In Section 2, a comprehensive review of the related literature is presented. Section 3 elaborates the proposed framework of DAO. Section 4 presents the experimental studies, along with the analysis of the results. Finally, Section 5 concludes the paper.

2. Related work

2.1. Constraint handling techniques

Categorization of CHTs in CMOEAs can be based on the size of the employed population, resulting in two distinct classifications: single-population CHTs and multi-population CHTs.

The single-population CHTs can be subdivided into two distinct categories: sort-based CHTs and tradeoff-based CHTs. For sort-based CHTs, the individuals of the population are compared according to the objective values and/or the degree of the constraint violation. To be specific, when comparing two solutions, we must take into account three scenarios in the constraint domination principle (CDP) [5]. If both are not feasible, we should choose the one that has a lower degree of constraint violation. If only one is feasible, then we should choose the feasible one. If both are feasible, the one with smaller values of the objective functions will be chosen. Different from CDP, some infeasible solutions in Epsilon CHT are regarded as feasible by relaxing constraints [14,44,45]. For tradeoff-based CHTs, the process of evolution takes into account both the optimization of objective functions and the constraint satisfaction, simultaneously. One approach, for instance, is to use penalty-based CHTs where the constraint violation is incorporated into objective functions by a penalty coefficient [4,12]. In another approach, i.e., $(M+1)$ -objective based CHT, the minimization of the constraint violation is considered as an additional optimization objective [13,28,32]. For sort-based CHTs, the individuals of the population are easy to get trapped into the local optimum since the preferred feasible solutions. For tradeoff-based CHTs, the existence of too many infeasible solutions will lead to a lack of exploration of the feasible region. To further improve the performance of the algorithm, Liu et al. [21], Fan et al. [10] and Tian et al. [36] take advantage of the different CHTs by employing different CHTs at different search stages.

In order to overcome the limitations of a single population, multi-population CHTs emerged as the times required. For example, in [37], a dual-population collaborative optimization framework was proposed, with the main population conducting the constrained search and the secondary population adopting an unconstrained search form; Wang et al. [38] divided a constrained multi-objective optimization problem into multiple constrained single-objective optimization problems to improve algorithm's performance; Zhang et al. [27] proposed a dual-population evolutionary algorithm, where the first population uses an improved penalty function to retain some infeasible solutions with good objective values to promote algorithm convergence, while the second population tends to retain feasible solution individuals; Li et al. [18] used two archives to balance convergence, diversity, and feasibility, with one archive focused on convergence to drive the population towards the PF and the other archive focused on diversity to improve the distribution performance of the population. These multi-population CMOEAs have better comprehensive performance than single-population methods and have gradually become a research trend for solving complex constrained multi-objective optimization problems.

2.2. CMOEAs based on auxiliary optimization

In multi-population CMOEAs, one way is to facilitate the solving of the CMOPs by means of the auxiliary optimization problems. Then, the CMOPs and the auxiliary problems are solved together.

One of the popular used auxiliary optimization problems is the unconstrained optimization problem. By ignoring the constraints, the convergence speed of the population can be accelerated [30,41]. For example, Zhang et al. [37] proposed a dual-population evolutionary framework, where the first population is used to solve the original CMOP, while the second population tends to the unconstrained optimization problem. In [46], by constructing the constraint-centric problem and objective-centric problem, and then solving them using Population 1 and Population 2, respectively. Tian et al. [35] Proposed a tri-population framework, which the first population evolve for the original CMOP, the second population and the third population evolve for the unconstrained MOP and the constraint-relaxed problem, respectively. Qiao et al. [31] design a auxiliary task with dynamically reduces the constraint boundary, the population can approaching the feasible boundary via knowledge transfer in the later stage.

Another auxiliary optimization problem is the $(M+1)$ -objective optimization problem. By treating the constraint violation minimization as an additional optimization objective, the diversity of the population is enhanced and the population is easy to jump out of the local optimum. For example, the population in [11] is separated into populations with feasible solutions and populations with infeasible solutions. The population with infeasible solutions is updated according to the $(M+1)$ -objective functions. In [39], the evolution population is responsible for the $(M+1)$ -objective optimization problem. The feasible solutions are saved to an external archive and evolve together with the evolution population.

Besides the two auxiliary optimizations mentioned above, an arbitrary number of auxiliary optimization tasks are constructed by different strategies in [26]. The reinforcement learning techniques are utilized according to the population state to choose the most suitable auxiliary task during the evolutionary process.

As mentioned above, CMOEAs based on auxiliary optimization have become a research trend for solving multi-objective optimization problems with complex constraints and have received extensive attention.

3. Proposed algorithm

3.1. Motivation

In current CMOEAs based on auxiliary optimization, the auxiliary optimization problem is usually framed by one simplified form of the original CMOP and the strategy of bi-population co-evolution is used to address the simplified problem and the original CMOP. The primary population plays the role of addressing the core CMOP, while the auxiliary population is dedicated to resolving the less complex problem. When the auxiliary optimization problem is the unconstrained problem. It is obvious that the population for tackling the easy problem will converge quickly. However, when the auxiliary optimization problem is unconstrained, it becomes evident that the population targeting the simpler problem converges quickly. Due to differing convergence speeds, the two populations gradually separate in the objective space. Simultaneously, individuals in the auxiliary population encounter challenges evolving towards feasible solutions, as the unconstrained problem neglects constraint satisfaction. In such a scenario, the auxiliary population struggles to contribute effectively to solving the CMOP. To solve this issue, we get help from another auxiliary optimization, i.e., the $(M+1)$ -objective optimization. As illustrated in Fig. 1, both the PF of the unconstrained optimization problem and the original CMOP are the boundary PF of the $(M+1)$ -objective optimization problem. The $(M+1)$ -objective optimization problem can build a bridge between the original constrained multi-objective optimization problem and the unconstrained optimization problem. Therefore, the two auxiliary optimization problems can get together to facilitate the solving of the original CMOP. Motivated by this inspiration, we propose a novel constrained multi-objective evolutionary algorithm assisted by double auxiliary optimization, called DAO.

3.2. Tri-population co-evolution strategy

Through a proposed tri-population co-evolution strategy, DAO addresses the core CMOP and the double auxiliary optimization problems in a simultaneous manner. The specific strategy is illustrated as Fig. 2 shows. Population1 solves the original CMOP by constrained search. Population2 handles the $(M+1)$ -objective auxiliary optimization problem by a proposed $(M+1)$ -objective environmental selection strategy, which will be illustrated in Section 3.3. Population3 tackles the unconstrained auxiliary optimization problem by unconstrained search. As illustrated in Section 2.2, the unconstrained auxiliary optimization and the $(M+1)$ -objective auxiliary optimization can enhance the convergence and the diversity of the algorithm respectively. Therefore, the proposed tri-population co-evolution strategy realizes a good balance of the feasibility, convergence and distribution. During the entire evolution process, Population3 is responsible for accelerating the convergence, Population2 helps Population1 jump out of the local optimum, and Population1 conducts deep mining of the optimal feasible region.

Specifically, one population is randomly selected to produce offspring at each generation. The generated offspring individuals from one population are used to update the whole population simultaneously. Different from traditional parallel evolution where all the population generate offspring in one generation [37], only one population is randomly selected to produce offspring individuals in the proposed tri-population co-evolution strategy. In this way, fewer individuals of the population will be replaced by the offspring at each generation iteration. Compared to traditional parallel evolution, the exploitation capability of the potential optimal solution region is enhanced.

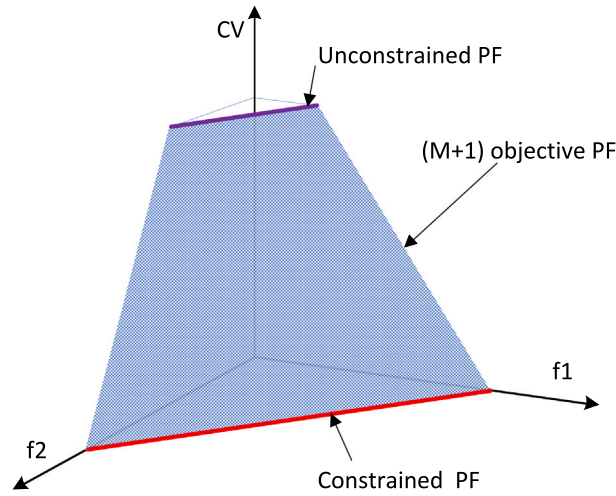


Fig. 1. Schematic diagram of the PF for the three problems.

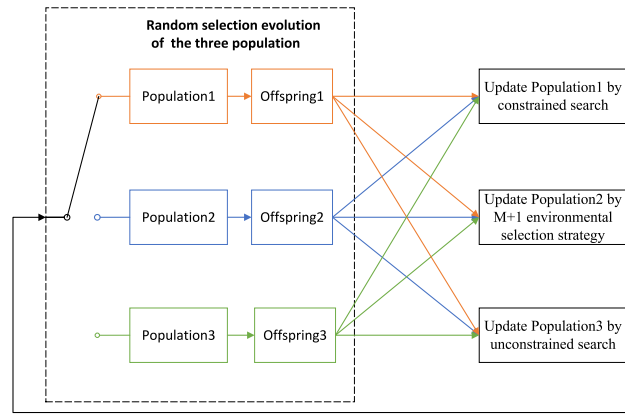


Fig. 2. Strategy of the tri-population co-evolution.

3.3. $(M+1)$ -objective environmental selection

When the number of individuals in population $\mathcal{A}$ is larger than the population number N_p , N_p individuals of $\mathcal{A}$ should be chosen to construct a new population $\mathcal{A}'$ by environmental selection. Algorithm 1 outlines the progress of the proposed $(M+1)$ -objective environmental selection. Specifically, the non-domination sorting is performed for individuals of $\mathcal{A}$ according to the $(M+1)$ optimization objective values. The individuals characterized by the number i Pareto rank collectively constitute the set $\mathcal{F}_i$ (line 2). After that, the individuals in $\mathcal{F}_1, \dots, \mathcal{F}_{i-1}$ are put into $\mathcal{A}'$ until $|\mathcal{A}'| + |\mathcal{F}_i|$ is larger than N_p (lines 3–7). Next, the non-domination sorting is also used for individuals of $\mathcal{F}_i$ according to the opposite values of M objective functions. The individuals assigned the i th Pareto rank are categorized within the set $\mathcal{BF}_i$ (line 8). Furthermore, the individuals in $\mathcal{BF}_1, \dots, \mathcal{BF}_{i-1}$ are put into $\mathcal{A}'$ until $|\mathcal{A}'| + |\mathcal{BF}_i|$ is larger than N_p (lines 9–12). After that, the crowding distance is used to cut out the individuals of $\mathcal{BF}_i$ for ensuring the number of individuals in $\mathcal{BF}_i$ is $N_p - |\mathcal{A}'|$. In the end, $\mathcal{BF}_i$ is added to $\mathcal{A}'$, which is the final output (lines 13–16).

In $(M+1)$ -objective environmental selection, the non-domination sorting achieved by the opposite M optimization objective values is processed after the $(M+1)$ -objective based non-domination sorting. The schematic of this progress is shown in Fig. 3. As Fig. 3 shows, we only select individuals close to the boundary of feasible and infeasible solutions when the $(M+1)$ -objective population is converged to the First Pareto rank. In this way, the boundary between feasible and infeasible solutions could be further explored, which helps obtain feasible solutions in different regions and further promotes the solution of original CMOPs.

3.4. Algorithm framework

Algorithm 2 presents the general framework of DAO. During the initialization phase, we initially assign $G = 0$ as the generation count and proceed to randomly formulate the first population P_1 , followed by the second population P_2 and the third population

Algorithm 1 (M+1)-objective Environmental Selection**Require:** Set $\mathcal{A}$;**Ensure:** Set $\mathcal{A}'$;

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1: if  $|\mathcal{A}| > N_p$  then
2:    $\{F_1, F_2, \dots\} \leftarrow$  Perform the non-domination-sorting on  $\mathcal{A}$  according to the  $(M+1)$  optimization objective values;
3:    $\mathcal{A}' \leftarrow \emptyset$  and  $l \leftarrow 1$ ;
4:   repeat
5:      $\mathcal{A}' \leftarrow \mathcal{A}' \cup F_l$ ;
6:      $l \leftarrow l + 1$ ;
7:   until  $|\mathcal{A}'| + |F_l| > N_p$ 
8:    $\{BF_1, BF_2, \dots\} \leftarrow$  Perform the non-domination-sorting on  $F_l$  according to the opposite  $M$  optimization objective values;
9:   repeat
10:     $\mathcal{A}' \leftarrow \mathcal{A}' \cup BF_i$ ;
11:     $i \leftarrow i + 1$ ;
12:  until  $|\mathcal{A}'| + |BF_i| > N_p$ 
13:   $CrowDis \leftarrow \text{CrowdingDistance}(BF_i)$ ;
14:  Sort  $CrowDis$  in descending order and select the top  $N_p - |\mathcal{A}'|$  of  $BF_i$  into  $\mathcal{A}'$ ;
15: else
16:    $\mathcal{A}' \leftarrow \mathcal{A}$ ;
17: end if

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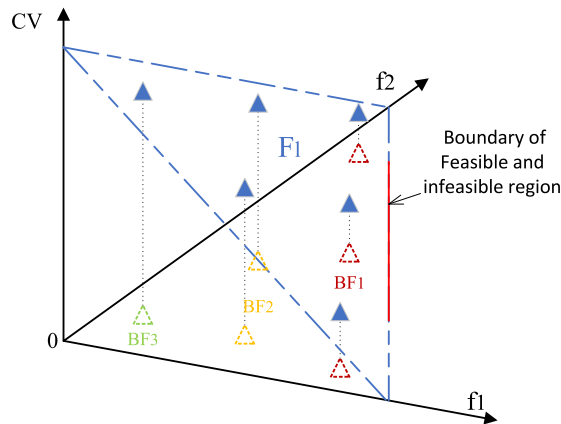


Fig. 3. Schematic diagram of non-domination-sorting based on the opposite values of the original M optimization objective.

P_3 (specified in lines 1–2). As the primary iteration unfolds, a singular population $\mathcal{P}$ is arbitrarily chosen from P_1 , P_2 , and P_3 (as demonstrated in lines 5–6) to generate the offspring $\mathcal{P}'$. After that, The offspring are used to update the three populations P_1 , P_2 and P_3 (line 7–9). Specifically, P_1 is updated based on the constrained search, P_2 is updated by the $(M+1)$ -objective environmental selection in Algorithm 1. P_3 is updated based on the unconstrained search. After reaching the number of iterations, the algorithm terminates. Finally, P_1 is output.

Fig. 4 shows the schematic of the search process of DAO. Initially, all three populations are located far away from the true PF. After that, the first population obtains feasible solutions in the local optimum, the third population moves towards the unconstrained PF with the assistance of unconstrained optimization. Solutions in the second population are distributed between the first population and the third population. Subsequently, the third population crosses the feasible regions containing the true PF. Meanwhile, the second population acquires feasible solutions in true PF. Then, the first population generates more feasible solutions in the same feasible region. After that, the second population obtains some solutions in other feasible regions. The boundary between feasible and infeasible solutions is deeply explored with the help of $(M+1)$ -objective environmental selection. Finally, feasible solutions with good convergence and distribution are realized in the first population.

4. Experimental studies

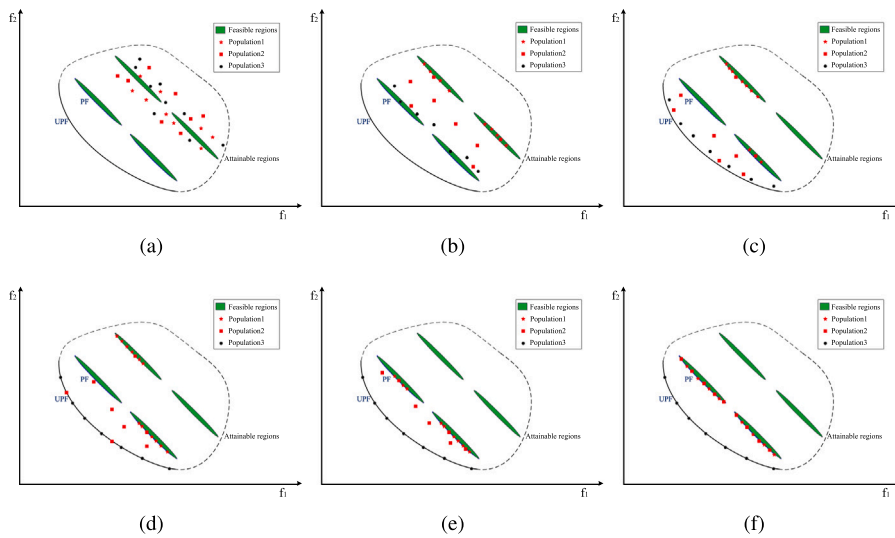
Four advanced algorithms: CCMO [37], NRC2 [20], ToP [21], and MOEA/D-2WA [13] are used to test the performance of DAO on three benchmark test sets, i.e., the DOC test functions [22], the DC-DTLZ test functions [18], and the LIRC-MOP test functions [9], and two real-world problems, i.e., the bi-source compressed-air pipeline optimization problem [42] and disk brake optimization problem [43]. NRC2, CCMO, MOEA/D-2WA, and ToP are four recent proposed CMOEAs. For the four algorithms,

Algorithm 2 General Framework of DAO**Require:** G_{max} ;**Ensure:** Set P_1 ;

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1: Initialize the iteration number  $G = 0$ ;
2: Generate randomly the first population  $P_1$ , the second population  $P_2$  and the third population  $P_3$ ;
3: while  $G < G_{max}$  do
4:    $P' = null$ ;
5:    $P \leftarrow$  Randomly select one population from  $P_1$ ,  $P_2$  and  $P_3$ ;
6:    $P' \leftarrow$  Perform evolutionary operators on  $P$ ;
7:    $P_1 \leftarrow$  Perform the environmental selection on  $P_1 \cup P'$  based on constrained search;
8:    $P_2 \leftarrow$  Perform the  $(M + 1)$ -objective environmental selection on  $P_3 \cup P'$ ;
9:    $P_3 \leftarrow$  Perform the environmental selection on  $P_3 \cup P'$  based on unconstrained search;
10: end while
11:  $G \leftarrow G + 1$ ;

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**Fig. 4.** Schematic diagram of the search process of DAO.

CCMO is a two-population algorithm assisted by unconstrained optimization. MOEA/D-2WA is a CMOEA based on $(M+1)$ -objective auxiliary optimization. ToP is a two-phase CMOEA while NRC2 applies a two-archive mechanism.

4.1. Experiments on benchmark test sets

Within this specific subsection, DAO is compared with four other competitors, namely CCMO, MOEA/D-2WA, NRC2, and ToP, on the above three test sets. The specific settings of the experiment are as follows:

1. **Benchmark Test Sets:** The LIRCMOP and DOC test sets consist of 14 and 9 test functions, respectively. Among them, LIRCMOP13, LIRCMOP14, DOC7, DOC8 and DOC9 have three optimization objectives. The remaining test functions a dual optimization objective structure. For the DOC and LIRCMOP test sets, the count of decision variables stands at 30 and 10 respectively. Meanwhile, the test set of DC-DTLZ encompasses a compilation of six test functions: specifically, DC1-DTLZ1, DC2-DTLZ1, DC3-DTLZ1, DC1-DTLZ3, DC2-DTLZ3, and DC3-DTLZ3. Each of them has three optimization objectives. For DC3-DTLZ1, DC2-DTLZ1, and DC1-DTLZ1, the number of decision variables is seven. For the other three problems, i.e., DC3-DTLZ3, DC2-DTLZ3, and DC1-DTLZ3, the number is twelve.
2. **Algorithm parameters:** For DAO, there are no specific parameter settings that need to be set. The four competitors (CCMO, ToP, NRC2, and MOEA/D-2WA) have recommended parameter settings that should be followed, as described in their respective papers [8,13,18,37].
3. **Evolutionary operators:** As adopted in [37], the evolutionary operators used for the DOC and the LIRCMOP test functions are “DE/rand/1” mutation and binomial crossover. The scale factor is set to 0.5. The crossover rate is set to 1.0. In terms of the

Table 1
Results of the IGD metric for the five algorithms on each test function.

Problem	CCMO	NRC2	ToP	MOEAD2WA	DAO
DOC1	5.8566e−3 (6.20e−4) ≈	1.0314e+0 (1.14e+0) −	5.8882e−3 (2.64e−4) −	8.5369e−1 (1.32e+0) −	5.5171e−3 (4.87e−4)
DOC2	3.3683e−2 (9.70e−2) −	2.4093e−1 (1.84e−1) −	4.6265e−1 (5.01e−2) −	NaN (NaN)	4.0098e−3 (5.76e−4)
DOC3	5.8715e+2 (4.89e+2) ≈	5.6606e+2 (2.68e+2) ≈	1.7786e+2 (1.81e+2) +	NaN (NaN)	5.2023e+2 (4.34e+2)
DOC4	2.4467e−2 (5.51e−3) −	3.8744e−1 (4.83e−1) −	7.2330e−2 (7.60e−2) ≈	1.5915e+0 (1.46e+0) −	1.8613e−2 (2.02e−3)
DOC5	7.7225e+0 (3.16e+1) ≈	NaN (NaN)	3.0976e+1 (4.48e+1) −	NaN (NaN)	2.9499e−2 (1.43e−2)
DOC6	3.3788e−2 (1.29e−1) −	6.6451e−1 (6.79e−1) −	2.0688e+0 (1.10e+0) −	2.6960e+0 (2.47e+0) −	3.0983e−3 (3.44e−4)
DOC7	2.4986e−3 (9.91e−5) ≈	3.5924e+0 (2.19e+0) −	2.8819e−1 (2.93e−1) −	2.5143e+0 (1.73e+0) −	4.6536e−2 (2.15e−1)
DOC8	7.2406e−2 (3.72e−3) +	6.0516e+1 (5.10e+1) −	1.5756e+1 (1.05e+1) −	6.5516e+1 (4.04e+1) −	8.2497e−2 (6.54e−3)
DOC9	8.2915e−2 (1.14e−2) ≈	4.1762e−2 (5.32e−3) +	1.7352e−1 (3.14e−2) −	3.7410e−1 (5.45e−2) −	8.7090e−2 (9.65e−3)
LIRC MOP1	5.0544e−2 (4.21e−2) −	1.0542e−2 (3.16e−3) +	2.2439e−1 (1.19e−1) −	7.3092e−2 (1.48e−2) −	1.1493e−2 (2.21e−3)
LIRC MOP2	5.3371e−2 (2.84e−2) −	9.0451e−3 (1.20e−3) ≈	1.1024e−1 (7.68e−2) −	7.9411e−2 (2.09e−2) −	8.4587e−3 (1.13e−3)
LIRC MOP3	1.0456e−1 (4.25e−2) −	1.0904e−2 (7.88e−3) ≈	3.2008e−1 (3.51e−2) −	8.8361e−2 (2.19e−2) −	5.9861e−3 (2.56e−3)
LIRC MOP4	1.1008e−1 (4.32e−2) −	1.2050e−2 (1.30e−2) ≈	3.0513e−1 (4.77e−2) −	1.2527e−1 (2.69e−2) −	4.9126e−3 (1.15e−3)
LIRC MOP5	5.5443e−3 (2.00e−4) −	5.2805e−1 (5.69e−1) −	5.3745e−1 (5.77e−1) −	2.7100e−1 (2.39e−2) −	5.3282e−3 (1.66e−4)
LIRC MOP6	5.5652e−3 (1.46e−4) −	4.0618e−1 (2.76e−1) −	3.3808e−1 (2.55e−1) −	2.9422e−1 (9.91e−2) −	5.3513e−3 (2.03e−4)
LIRC MOP7	7.1790e−3 (1.88e−4) ≈	3.0060e−2 (2.43e−2) −	9.5041e−2 (3.42e−1) −	1.0795e−1 (1.76e−2) −	7.1089e−3 (1.89e−4)
LIRC MOP8	7.1544e−3 (2.09e−4) ≈	2.5067e−2 (2.15e−2) −	1.5226e−1 (2.86e−1) −	1.7123e−1 (4.04e−2) −	7.0883e−3 (2.25e−4)
LIRC MOP9	2.8699e−3 (1.16e−4) +	1.0635e−1 (2.28e−2) −	3.7073e−1 (4.07e−2) −	1.6990e−1 (4.79e−2) −	3.0774e−3 (1.15e−4)
LIRC MOP10	5.0179e−3 (2.27e−4) −	1.5566e−2 (1.56e−2) −	5.9295e−3 (3.26e−4) −	2.1193e−2 (3.04e−2) −	4.8675e−3 (1.86e−4)
LIRC MOP11	2.4497e−3 (3.96e−5) ≈	2.5396e−1 (1.24e−1) −	2.4184e−1 (6.44e−2) −	1.6474e−1 (1.72e−3) −	2.4628e−3 (5.81e−5)
LIRC MOP12	3.1606e−3 (1.83e−4) ≈	4.8925e−2 (2.26e−2) −	1.1835e−1 (7.28e−2) −	1.4086e−1 (6.72e−2) −	3.2018e−3 (2.56e−4)
LIRC MOP13	1.1236e−1 (1.93e−3) ≈	3.4916e−1 (5.08e−1) −	1.3122e−1 (4.66e−3) −	1.4333e−1 (2.47e−1) −	1.1178e−1 (1.65e−3)
LIRC MOP14	1.0112e−1 (1.03e−3) ≈	3.9689e−1 (5.18e−1) −	1.2073e−1 (2.83e−3) −	1.4382e−1 (2.38e−1) −	1.0137e−1 (1.19e−3)
DC1DTLZ1	1.1429e−2 (7.73e−5) ≈	1.1330e−2 (1.20e−4) +	2.2000e−2 (6.30e−3) −	2.6947e−2 (5.54e−3) −	1.1408e−2 (6.50e−5)
DC1DTLZ3	3.3754e−2 (3.06e−4) ≈	3.3368e−2 (3.58e−4) +	1.4188e+0 (2.25e+0) −	4.5103e−2 (4.98e−5) −	3.3827e−2 (2.51e−4)
DC2DTLZ1	2.0135e−2 (1.59e−4) ≈	2.5745e−2 (2.98e−2) −	NaN (NaN)	2.0580e−2 (3.91e−5) −	2.0083e−2 (1.45e−4)

(continued on next page)

DC-DTLZ test functions, the evolutionary operators include the polynomial mutation and simulated binary crossover. Both the mutation distribution index and the cross distribution index are fixed at 20. The mutation probability and the crossover probability are defined as $1/N$ and 1.0, respectively.

Table 1 (continued).

DC2DTLZ3	5.3126e-2 (6.47e-4) $\approx$	5.4374e-1 (1.10e-1) –	NaN (NaN)	3.9278e-1 (2.44e-1) –	5.3085e-2 (5.18e-4)
DC3DTLZ1	6.8449e-3 (6.30e-5) $\approx$	9.9968e-2 (1.04e-1) –	2.2785e+0 (2.00e+0) –	2.0283e-2 (6.69e-3) –	6.8530e-3 (6.37e-5)
DC3DTLZ3	2.0056e-2 (2.66e-4) $\approx$	1.3443e+0 (5.68e-1) –	7.9605e+0 (3.93e+0) –	3.8718e-1 (2.83e-1) –	2.0159e-2 (3.77e-4)
+ / – / $\approx$	2/10/17	4/20/4	1/25/1	0/26/0	

–, +, and $\approx$ indicate the algorithm performs significantly worse than, better than, and equivalent to DAO, respectively.

4. Population size and terminal condition: The population size NP is set to 100. The algorithms stop 100,000 function evaluations for the LIRCMOP test set and 300,000 for the DOC and DC-DTLZ.

After performing 30 independent runs, the results are evaluated using two performance indicators: the inverted generational distance (IGD) [2] and the hypervolume (HV) [49] indicators. Additionally, the Wilcoxon rank-sum test is used to determine whether there is a statistically significant difference in performance between two algorithms. In the tables, NaN means the corresponding algorithm is unable to obtain feasible solutions every time.

Table 1 provides an overview of the mean values and standard deviations of the Inverted Generational Distance (IGD) metric for each test problem. The algorithm that exhibits the best performance is emphasized by being presented in boldface. Table 1 also shows the results of the Wilcoxon rank-sum test between DAO and each compared algorithm. From the results, it can be observed that DAO achieves the best performance on 16 out of 29 test problems, while CCMO and NRC2 obtained the best results on eight and four test problems, respectively. ToP only achieves the best results on one test problem, and MOEA/D-2WA does not perform as well as the other algorithms. Furthermore, the results of the Wilcoxon rank-sum test illustrate that DAO exhibits statistically significant superior performance compared to CCMO, NRC2, ToP, and MOEA/D-2WA on 10, 20, 26, and 26 test problems respectively. Conversely, DAO exhibits statistically significant inferior performance compared to these algorithms on only two, four, one, and zero test problems respectively.

Table 2 shows the mean values and standard deviations of the HV metric for each algorithm on each test function. The best values of the mean HV indicator are highlighted in boldface. Similar to the IGD indicator results, DAO outperforms the other algorithms on a majority of the test problems, achieving the best results on 15 out of 29 test problems. CCMO achieves the best results on 12 test problems, while NRC2 and ToP do not achieve the best results on any test problem. MOEA/D-2WA achieves the best performance on only one test problem. Furthermore, the tabulated Wilcoxon rank-sum test results affirm that DAO consistently exhibits superior or comparable performance in comparison to the other algorithms on the majority of the test functions.

To showcase the superiority of DAO, Fig. 5 illustrates the typical results for the five algorithms on DOC4, where the constrained Pareto Front (PF) is comprised of scattered points. As depicted in Fig. 5, NRC2, ToP, and MOEA/D-2WA gets trapped in local optima, resulting in insufficient convergence. On the other hand, DAO and CCMO demonstrate better performance by nearly obtaining all solutions of the constrained PF. However, compared to DAO, CCMO exhibits poor convergence for some points. Consequently, DAO achieves the best performance on DOC4.

The results indicate that DAO achieves good distribution and convergence, which is attributed to the combination of the first, second, and third populations focusing on feasible regions, objective space, and the whole space, respectively. However, CCMO and NRC2 exhibit poor performance on some test functions despite using two archives or populations to maintain diversity and convergence separately. The performance of ToP is unstable due to insufficient convergence in the early stages, leading to large variances in performance on many test functions. With the gradual reduction of population diversity, MOEA/D-2WA is more likely to get trapped in local optima. Therefore, DAO consistently showcases superior performance in contrast to the other peer algorithms.

4.2. Comparisons between DAO and its variants

In this subsection, ablation experiments are conducted to demonstrate the effectiveness of the proposed algorithm. As described in Section 3.2, offspring individuals are generated from only one population in the proposed tri-population co-evolution strategy. As a comparison, DAOv1 is designed based on all the population generating offspring in one generation. Furthermore, as described in Section 3.3, the non-domination sorting achieved by the opposite M optimization objective values is processed after the $(M+1)$ -objective based non-domination sorting. As a comparison, DAOv2 is designed without the non-domination sorting achieved by the opposite M optimization objective values, which means DAOv2 excludes lines 8–12 in Algorithm 1. In this experimental studies, DOC test problems are utilized to assess the performance of DAO and its variants. The mean value and standard deviation of the IGD indicator obtained by DAO and its variants are presented in Table 3. It can be seen that DAO performs significantly better or equivalent performance compared to DAOv1 and DAOv2 on most test functions. It indicates that the tri-population co-evolution strategy and $(M+1)$ -objective environment selection in DAO is effectiveness.

Table 2
Results of the HV metric for the five algorithms on each test function.

Problem	CCMO	NRC2	ToP	MOEAD2WA	DAO
DOC1	3.4446e−1 (7.42e−4) −	1.2339e−1 (1.54e−1) −	3.4455e−1 (4.14e−4) −	9.1993e−2 (6.77e−2) −	3.4497e−1 (5.82e−4)
DOC2	5.9712e−1 (7.24e−2) −	3.9650e−1 (1.41e−1) −	2.5949e−1 (5.45e−2) −	NaN (NaN)	6.2098e−1 (9.10e−4)
DOC3	5.3051e−2 (1.21e−1) ≈	0.0000e+0 (0.00e+0) ≈	6.7590e−3 (3.31e−2) ≈	NaN (NaN)	3.9509e−2 (1.01e−1)
DOC4	5.3486e−1 (6.25e−3) −	3.0448e−1 (1.95e−1) −	4.8448e−1 (6.59e−2) ≈	1.0475e−2 (2.30e−2) −	5.4175e−1 (2.46e−3)
DOC5	4.4417e−1 (1.18e−1) −	NaN (NaN)	2.5348e−1 (1.99e−1) −	NaN (NaN)	4.8173e−1 (1.15e−2)
DOC6	4.7572e−1 (1.04e−1) −	1.6073e−1 (1.91e−1) −	2.2296e−4 (1.09e−3) −	3.3668e−3 (9.21e−3) −	5.2860e−1 (5.79e−3)
DOC7	5.3043e−1 (9.96e−3) +	1.4916e−2 (7.31e−2) −	2.3604e−1 (1.63e−1) −	1.4578e−2 (4.44e−2) −	5.1368e−1 (1.11e−1)
DOC8	7.8711e−1 (4.91e−3) +	0.0000e+0 (0.00e+0) −	0.0000e+0 (0.00e+0) −	0.0000e+0 (0.00e+0) −	7.7536e−1 (8.23e−3)
DOC9	0.0000e+0 (0.00e+0) ≈	0.0000e+0 (0.00e+0) ≈	0.0000e+0 (0.00e+0) ≈	0.0000e+0 (0.00e+0) ≈	0.0000e+0 (0.00e+0)
LIRC MOP1	2.0877e−1 (2.25e−2) −	2.3452e−1 (2.71e−3) ≈	1.4576e−1 (4.44e−2) −	2.0228e−1 (5.92e−3) −	2.3516e−1 (1.24e−3)
LIRC MOP2	3.2537e−1 (1.96e−2) −	3.5728e−1 (1.39e−3) −	2.9903e−1 (3.33e−2) −	3.2542e−1 (7.80e−3) −	3.5816e−1 (7.35e−4)
LIRC MOP3	1.6328e−1 (1.73e−2) −	2.0439e−1 (3.44e−3) ≈	9.6427e−2 (1.12e−2) −	1.7266e−1 (8.65e−3) −	2.0520e−1 (1.85e−3)
LIRC MOP4	2.6648e−1 (2.01e−2) −	3.1223e−1 (6.62e−3) ≈	1.8805e−1 (2.46e−2) −	2.7333e−1 (7.02e−3) −	3.1452e−1 (9.28e−4)
LIRC MOP5	2.9187e−1 (1.13e−4) −	1.5229e−1 (1.34e−1) −	1.5582e−1 (1.47e−1) −	1.7087e−1 (1.33e−2) −	2.9200e−1 (9.15e−5)
LIRC MOP6	1.9709e−1 (8.38e−5) −	1.0626e−1 (4.47e−2) −	9.1390e−2 (5.09e−2) −	1.3368e−1 (1.34e−2) −	1.9721e−1 (1.09e−4)
LIRC MOP7	2.9456e−1 (8.74e−5) ≈	2.8088e−1 (1.13e−2) −	2.7504e−1 (6.15e−2) −	2.5418e−1 (6.41e−3) −	2.9460e−1 (9.77e−5)
LIRC MOP8	2.9457e−1 (9.53e−5) ≈	2.8343e−1 (9.05e−3) −	2.6695e−1 (5.57e−2) −	2.3947e−1 (7.25e−3) −	2.9459e−1 (1.16e−4)
LIRC MOP9	5.6725e−1 (1.24e−4) +	5.2640e−1 (1.26e−2) −	4.5030e−1 (2.61e−2) −	5.1025e−1 (2.49e−2) −	5.6699e−1 (1.75e−4)
LIRC MOP10	7.0722e−1 (1.21e−4) ≈	7.0022e−1 (5.55e−3) −	7.0709e−1 (1.58e−4) −	6.9235e−1 (2.51e−2) −	7.0723e−1 (1.39e−4)
LIRC MOP11	6.9399e−1 (1.08e−5) +	5.3435e−1 (8.35e−2) −	5.4197e−1 (3.90e−2) −	6.8492e−1 (3.45e−3) −	6.9397e−1 (2.07e−5)
LIRC MOP12	6.2031e−1 (2.06e−5) +	6.0438e−1 (7.25e−3) −	5.6614e−1 (3.68e−2) −	5.9259e−1 (1.49e−2) −	6.2026e−1 (4.20e−5)
LIRC MOP13	5.2114e−1 (2.88e−3) −	4.3728e−1 (2.29e−1) −	5.0791e−1 (3.75e−3) −	5.3632e−1 (1.14e−1) +	5.2274e−1 (1.84e−3)
LIRC MOP14	5.4374e−1 (1.69e−3) ≈	4.0797e−1 (2.44e−1) −	5.2624e−1 (3.00e−3) −	5.3604e−1 (1.14e−1) −	5.4408e−1 (1.82e−3)
DC1DTLZ1	6.3305e−1 (5.69e−4) ≈	6.3275e−1 (8.94e−4) ≈	5.8042e−1 (2.43e−2) −	5.9820e−1 (1.86e−2) −	6.3270e−1 (8.34e−4)
DC1DTLZ3	4.7402e−1 (7.71e−4) ≈	4.7312e−1 (1.38e−3) ≈	1.7977e−1 (1.87e−1) −	4.7208e−1 (5.94e−4) −	4.7376e−1 (1.07e−3)

(continued on next page)

4.3. Experiments on real-world optimization problems

The objectives of the disk brake optimization problem (DISK) [40] are to simultaneously minimize the disk brake time (f_1) and the stopping time (f_2). DISK has four decision variables, i.e., the outer radius of the disk (x_1), the inner radius of the disk (x_2), the

Table 2 (continued).

DC2DTLZ1	8.4220e-1 (4.08e-4) $\approx$	8.2776e-1 (7.50e-2) –	NaN (NaN)	8.4144e-1 (4.05e-4) –	8.4210e-1 (6.64e-4)
DC2DTLZ3	5.6065e-1 (1.16e-3) +	3.6777e-2 (1.15e-1) –	NaN (NaN)	1.9127e-1 (2.65e-1) –	5.5961e-1 (1.40e-3)
DC3DTLZ1	5.3612e-1 (1.22e-3) $\approx$	3.0966e-1 (2.31e-1) –	1.2864e-2 (6.30e-2) –	5.0782e-1 (3.23e-2) –	5.3565e-1 (1.16e-3)
DC3DTLZ3	3.6772e-1 (1.26e-3) +	0.0000e+0 (0.00e+0) –	0.0000e+0 (0.00e+0) –	1.3732e-1 (1.81e-1) –	3.6586e-1 (2.09e-3)
+/-/ $\approx$	7/12/10	0/21/7	0/24/3	1/24/0	

–, +, and $\approx$ indicate the algorithm performs significantly worse than, better than, and equivalent to DAO, respectively.

Table 3

Performances of DAO and its variants on DOC test problems in terms of the IGD metric.

Problem	DAOv1	DAOv2	DAO
DOC1	5.6903e-3 (3.87-4) $\approx$	6.1383e-3 (6.05e-4) $\approx$	5.5171e-3 (4.87e-4)
DOC2	4.2253e-3 (6.54e-4) $\approx$	4.6668e-2 (1.09e-1) –	4.0098e-3 (5.76e-4)
DOC3	5.4346e+2 (2.74e+2) $\approx$	3.9925e+2 (4.01e+2) $\approx$	5.2023+2 (4.34e+2)
DOC4	1.8639e-2 (2.17e-3) $\approx$	3.3518e-2 (7.46e-3) –	1.8613e-2 (2.02e-3)
DOC5	2.9034e+1 (5.58e+1) –	1.6409+1 (4.47e+1) –	2.9499e-2 (1.43e-2)
DOC6	3.1083e-3 (2.67e-4) $\approx$	2.4328e-1 (3.03e-1) –	3.0983e-3 (3.44e-4)
DOC7	4.4661e+0 (2.47e+0) –	2.0961e-2 (7.02e-2) –	4.6536e-2 (2.15e-1)
DOC8	2.3033e+2 (1.46e+2) –	6.1313e+2 (1.05e+2) –	8.2497e-2 (6.54e-3)
DOC9	8.8925e-2 (1.41e-2) $\approx$	7.9368e-2 (9.63e-3) +	8.7090e-2 (9.65e-3)
+/-/ $\approx$	0/3/6	1/6/2	

–, +, and $\approx$ indicate the algorithm performs significantly worse than, better than, and equivalent to DAO, respectively.

engaging force (x_3), and the number of friction surfaces (x_4). The formulation of DISK is as follows:

$$\begin{aligned}
 &\text{minimize } f_1(x) = 9.8 \times 10^6 (x_1^2 - x_2^2) / x_4 x_3 (x_1^3 - x_2^3), \\
 &\text{minimize } f_2(x) = 4.9 \times 10^{-5} (x_1^2 - x_2^2) (x_4 - 1), \\
 &s.t. \begin{cases} 20 - (x_1 - x_2) \leq 0, \\ 2.5 (x_4 + 1) - 30 \leq 0, \\ x_3 / \pi (x_1^2 - x_2^2)^2 - 0.4 \leq 0, \\ 2.22 \times 10^{-3} x_3 (x_1^3 - x_2^3) / (x_1^2 - x_2^2)^2 - 1 \leq 0, \\ 900 - 2.66 \times 10^{-2} x_4 x_3 (x_1^3 - x_2^3) / (x_1^2 - x_2^2) \leq 0, \end{cases}
 \end{aligned} \tag{3}$$

where $75 \leq x_1 \leq 110$, $55 \leq x_2 \leq 80$, $1000 \leq x_3 \leq 3000$, and $2 \leq x_4 \leq 20$.

The second real-world problem under consideration is the bi-source compressed-air pipeline optimization problem (BSCPOP) [17]. The system structure of BSCPOP is as Fig. 6 shows. Minimizing the two output pressures of each source node $N_s = \{S_1, S_2\}$ are the two objectives of BSCPOP. $\vec{P}_D = \{P_{D_1}, P_{D_2}, P_{D_3}, P_{D_4}\}$ are the decision variables (i.e. four pressure values of the demand nodes $\{D_1, D_2, D_3, D_4\}$). The specific formulations of BSCPOP could be found in [42].

During the experimental phase, the parameter configurations for the algorithms remain consistent with those detailed in Section 4.1. For DISK, the algorithms will stop once the number of the function evaluations reaches 50,000. For BSCPOP, the number of the function evaluations is 200,000. In the experiments, the algorithm's performance is evaluated by the IGD indicator.

As shown in Table 4, DAO obtains the best performance on BSCPOP among the five algorithms. Furthermore, DAO is significantly better than the other four algorithms according to the results of the Wilcoxon rank-sum test. For DISK, NRC2 gets the best result. The performance of the proposed DAO is equivalent to NRC2. The other three algorithms show worse performance than DAO. DAO also shows competitive performance on real-world problems according to the experimental results.

5. Conclusion

This article proposed a constrained multi-objective evolutionary algorithm assisted by double auxiliary optimization, called DAO. Within the framework of DAO, a tri-population co-evolutionary strategy is meticulously designed. In this strategy, the initial population is primarily dedicated to exploring the feasible region. Simultaneously, the second population undertakes a dual responsibility of exploring both the constraint and objective spaces through the $(M+1)$ -objective optimization. Meanwhile, the third population focuses solely on exploring the objective space through unconstrained optimization. Moreover, the proposed $(M+1)$ -objective environmental selection strategy can fully explore the boundary of infeasible and feasible regions. The experimental results indicate the competitiveness of DAO. We will further investigate CMOEAs based on multi-auxiliary optimization including the other auxiliary optimization strategy and co-evolution strategy.

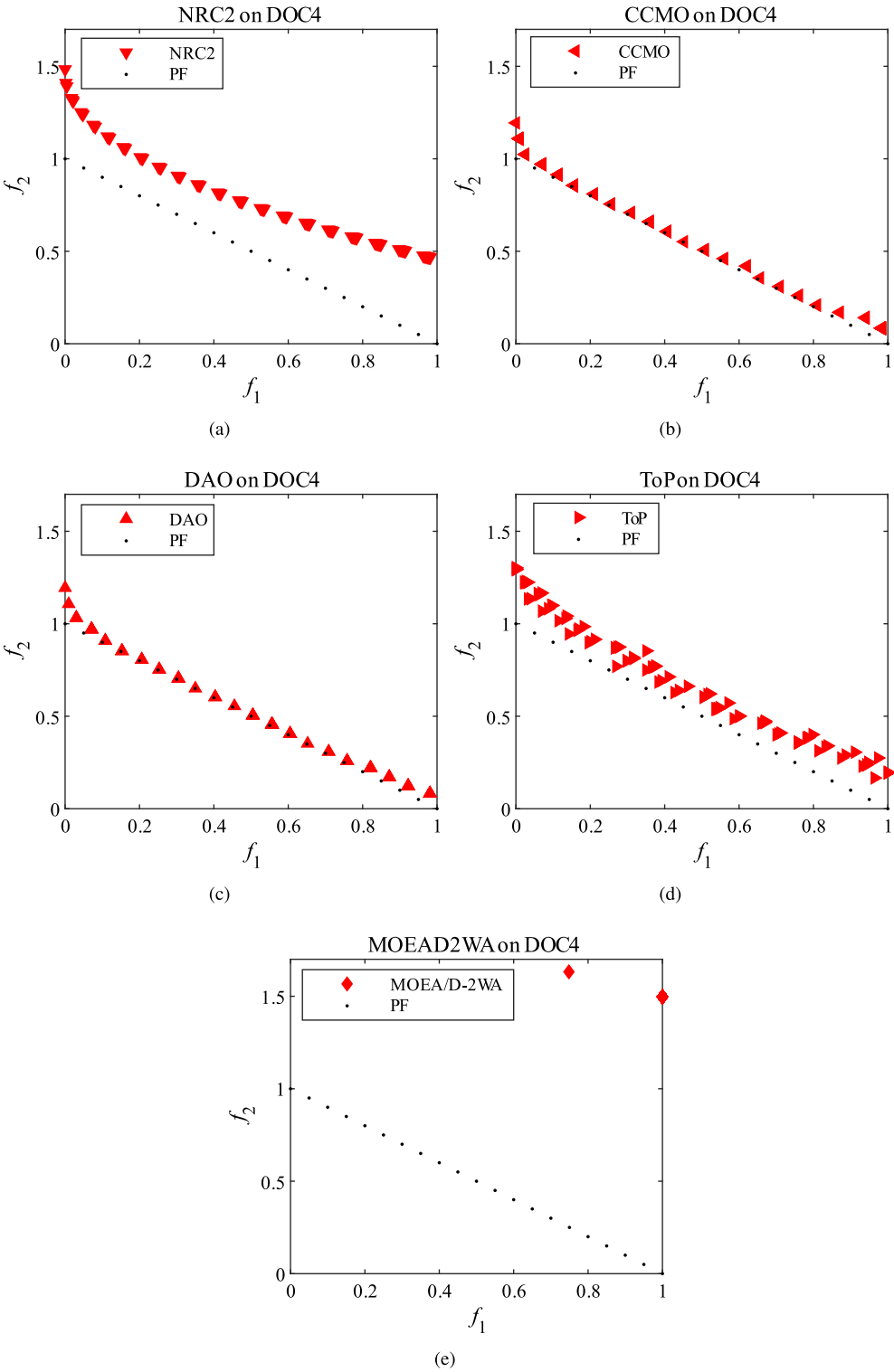


Fig. 5. Solutions achieved by each algorithm on DOC4.

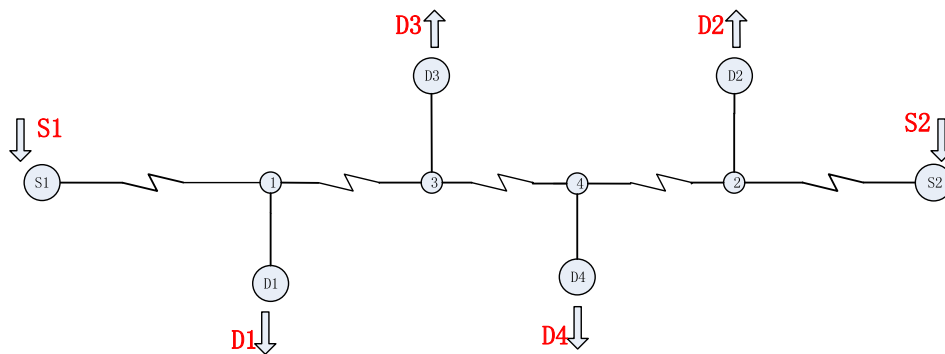


Fig. 6. Bi-source compressed-air pipeline optimization problem.

Table 4

Results of the IGD metric for NRC2, CCMO, ToP, MOEAD-2WA, and DAO on DISK and BSCPOP.

Problem	NRC2	CCMO	ToP	MOEAD2WA	DAO
BSCPOP	6.8658e-2 (7.11e-3) –	8.3297e-1 (1.77e+0) –	8.5010e-2 (4.72e-2) –	1.3869e+0 (1.95e-1) –	4.8893e-2 (1.28e-3)
DSK	3.4888e-2 (1.07e-3) ≈	1.4388e+0 (8.50e-3) –	4.0609e-2 (1.57e-3) –	1.1822e+0 (6.42e-3) –	3.5666e-2 (1.89e-3)
+ / – / ≈	0/1/1	0/2/0	0/2/0	0/2/0	

–, +, and ≈ indicate the algorithm performs significantly worse than, better than, and equivalent to DAO, respectively.

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References

- [1] M.-R. Chen, Y.-Z. Lu, G. Yang, Multiobjective optimization using population-based extremal optimization, *Neural Comput. Appl.* 17 (2) (2008) 101–109.
- [2] C.A.C. Coello, M.R. Sierra, A study of the parallelization of a coevolutionary multi-objective evolutionary algorithm, in: *Mexican International Conference on Artificial Intelligence*, Springer, 2004, pp. 688–697.
- [3] M. da Silva Maximiano, M.A. Vega-Rodríguez, J.A. Gómez-Pulido, J.M. Sánchez-Pérez, A new multiobjective artificial bee colony algorithm to solve a real-world frequency assignment problem, *Neural Comput. Appl.* 22 (7) (2013) 1447–1459.
- [4] R. Datta, K. Deb, A. Segev, A bi-objective hybrid constrained optimization (HyCon) method using a multi-objective and penalty function approach, in: *2017 IEEE Congress on Evolutionary Computation, CEC, 2017*, pp. 317–324.
- [5] K. Deb, A. Pratap, S. Agarwal, T. Meyarivan, A fast and elitist multiobjective genetic algorithm: NSGA-II, *IEEE Trans. Evol. Comput.* 6 (2) (2002) 182–197.
- [6] A. Demissie, W. Zhu, C.T. Belachew, A multi-objective optimization model for gas pipeline operations, *Comput. Chem. Eng.* 100 (2017) 94–103.
- [7] Z. Dong, X. Wang, L. Tang, MOEA/D with a self-adaptive weight vector adjustment strategy based on chain segmentation, *Inform. Sci.* 521 (2020) 209–230.
- [8] Z. Fan, W. Li, X. Cai, H. Li, C. Wei, Q. Zhang, K. Deb, E. Goodman, Push and pull search for solving constrained multi-objective optimization problems, *Swarm Evol. Comput.* 44 (2019) 665–679.
- [9] Z. Fan, W. Li, X. Cai, H. Li, C. Wei, Q. Zhang, K. Deb, E. Goodman, Difficulty adjustable and scalable constrained multiobjective test problem toolkit, *Evol. Comput.* 28 (3) (2020) 339–378.
- [10] Z. Fan, Z. Wang, W. Li, Y. Yuan, Y. You, Z. Yang, F. Sun, J. Ruan, Push and pull search embedded in an M2M framework for solving constrained multi-objective optimization problems, *Swarm Evol. Comput.* 54 (2020) 100651.
- [11] A. Isaacs, T. Ray, W. Smith, Blessings of maintaining infeasible solutions for constrained multi-objective optimization problems, in: *2008 IEEE Congress on Evolutionary Computation (IEEE World Congress on Computational Intelligence)*, 2008, pp. 2780–2787.
- [12] M.A. Jan, N. Tairan, R.A. Khanum, Threshold based dynamic and adaptive penalty functions for constrained multiobjective optimization, in: *International Conference on Artificial Intelligence, Modelling and Simulation, IEEE*, 2013, pp. 49–54.
- [13] R. Jiao, S. Zeng, C. Li, Y.-S. Ong, Two-type weight adjustments in MOEA/D for highly constrained many-objective optimization, *Inform. Sci.* 578 (2021) 592–614.
- [14] R. Jiao, S. Zeng, C. Li, S. Yang, Y.-S. Ong, Handling constrained many-objective optimization problems via problem transformation, *IEEE Trans. Cybern.* 51 (10) (2021) 4834–4847.
- [15] V.K. Kamboj, S. Bath, J. Dhillon, Multiobjective multiarea unit commitment using hybrid differential evolution algorithm considering import/export and tie-line constraints, *Neural Comput. Appl.* 28 (11) (2017) 3521–3536.
- [16] F. Karami, A.B. Dariane, A review and evaluation of multi and many-objective optimization: Methods and algorithms, *Glob. J. Ecol.* 7 (2) (2022) 104–119.
- [17] X. Kong, Y. Yang, Z. Lv, J. Zhao, R. Fu, A dynamic dual-population co-evolution multi-objective evolutionary algorithm for constrained multi-objective optimization problems, *Appl. Soft Comput.* 141 (2023) 110311.
- [18] K. Li, R. Chen, G. Fu, X. Yao, Two-archive evolutionary algorithm for constrained multiobjective optimization, *IEEE Trans. Evol. Comput.* 23 (2) (2019) 303–315.

- [19] F. Li, Y. Yang, Z. Shang, S. Li, H. Ouyang, Kriging-assisted indicator-based evolutionary algorithm for expensive multi-objective optimization, *Appl. Soft Comput.* 147 (2023) 110736.
- [20] Z.-Z. Liu, Y. Qin, W. Song, J. Zhang, K. Li, Multiobjective-based constraint-handling technique for evolutionary constrained multiobjective optimization: A new perspective, *IEEE Trans. Evol. Comput.* (2022) 1.
- [21] Z.-Z. Liu, Y. Wang, Handling constrained multiobjective optimization problems with constraints in both the decision and objective spaces, *IEEE Trans. Evol. Comput.* 23 (5) (2019) 870–884.
- [22] Z. Ma, Y. Wang, Evolutionary constrained multiobjective optimization: Test suite construction and performance comparisons, *IEEE Trans. Evol. Comput.* 23 (6) (2019) 972–986.
- [23] Z. Ma, Y. Wang, W. Song, A new fitness function with two rankings for evolutionary constrained multiobjective optimization, *IEEE Trans. Syst., Man, Cybern.: Syst.* 51 (8) (2021) 5005–5016.
- [24] H. Ma, H. Wei, Y. Tian, R. Cheng, X. Zhang, A multi-stage evolutionary algorithm for multi-objective optimization with complex constraints, *Inform. Sci.* 560 (2021) 68–91.
- [25] S.Z. Martinez, A.A. Montano, C.A.C. Coello, Constrained Multi-Objective Aerodynamic Shape Optimization Via Swarm Intelligence, *ACM*, 2014, pp. 81–88.
- [26] F. Ming, W. Gong, L. Gao, Adaptive auxiliary task selection for multitasking-assisted constrained multi-objective optimization [feature], *IEEE Comput. Intell. Mag.* 18 (2) (2023) 18–30.
- [27] M. Ming, A. Trivedi, R. Wang, D. Srinivasan, T. Zhang, A dual-population-based evolutionary algorithm for constrained multiobjective optimization, *IEEE Trans. Evol. Comput.* 25 (4) (2021) 739–753.
- [28] C. Peng, H.-L. Liu, F. Gu, An evolutionary algorithm with directed weights for constrained multi-objective optimization, *Appl. Soft Comput.* 60 (2017) 613–622.
- [29] S. Qian, Y. Ye, B. Jiang, J. Wang, Constrained multiobjective optimization algorithm based on immune system model, *IEEE Trans. Cybern.* 46 (9) (2016) 2056–2069.
- [30] K. Qiao, J. Liang, K. Yu, M. Wang, B. Qu, C. Yue, Y. Guo, A self-adaptive evolutionary multi-task based constrained multi-objective evolutionary algorithm, *IEEE Trans. Emerg. Top. Comput. Intell.* 7 (4) (2023) 1098–1112.
- [31] K. Qiao, K. Yu, B. Qu, J. Liang, H. Song, C. Yue, H. Lin, K.C. Tan, Dynamic auxiliary task-based evolutionary multitasking for constrained multi-objective optimization, *IEEE Trans. Evol. Comput.* (2022).
- [32] T. Ray, H.K. Singh, A. Isaacs, W. Smith, Infeasibility driven evolutionary algorithm for constrained optimization, in: *Constraint-Handling in Evolutionary Optimization*, Springer, 2009, pp. 145–165.
- [33] B. Tan, H. Ma, Y. Mei, M. Zhang, Evolutionary multi-objective optimization for web service location allocation problem, *IEEE Trans. Serv. Comput.* 14 (2) (2021) 458–471.
- [34] M.A. Tawhid, V. Savsani, Multi-objective sine-cosine algorithm (MO-SCA) for multi-objective engineering design problems, *Neural Comput. Appl.* 31 (2) (2019) 915–929.
- [35] Y. Tian, Z. Shi, Y. Zhang, L. Zhang, H. Zhang, X. Zhang, Solving optimal power flow problems via a constrained many-objective co-evolutionary algorithm, *Front. Energy Res.* 11 (2023) 1293193.
- [36] Y. Tian, Y. Zhang, Y. Su, X. Zhang, K.C. Tan, Y. Jin, Balancing objective optimization and constraint satisfaction in constrained evolutionary multiobjective optimization, *IEEE Trans. Cybern.* 52 (9) (2021) 9559–9572.
- [37] Y. Tian, T. Zhang, J. Xiao, X. Zhang, Y. Jin, A coevolutionary framework for constrained multiobjective optimization problems, *IEEE Trans. Evol. Comput.* 25 (1) (2021) 102–116.
- [38] J. Wang, G. Liang, J. Zhang, Cooperative differential evolution framework for constrained multiobjective optimization, *IEEE Trans. Cybern.* 49 (6) (2019) 2060–2072.
- [39] Y. Yang, P.-Q. Huang, X. Kong, J. Zhao, A constrained multi-objective evolutionary algorithm assisted by an additional objective function, *Appl. Soft Comput.* 132 (2023) 109904.
- [40] Y. Yang, J. Liu, S. Tan, A constrained multi-objective evolutionary algorithm based on decomposition and dynamic constraint-handling mechanism, *Appl. Soft Comput.* 89 (2020) 106104.
- [41] Y. Yang, J. Liu, S. Tan, A multi-objective evolutionary algorithm for steady-state constrained multi-objective optimization problems, *Appl. Soft Comput.* 101 (2021) 107042.
- [42] Y. Yang, J. Liu, S. Tan, A partition-based constrained multi-objective evolutionary algorithm, *Swarm Evol. Comput.* 66 (2021) 100940.
- [43] Y. Yang, J. Liu, S. Tan, Y. Liu, A multi-objective differential evolution algorithm based on domination and constraint-handling switching, *Inform. Sci.* 579 (2021) 796–813.
- [44] Y. Yang, J. Liu, S. Tan, H. Wang, A multi-objective differential evolutionary algorithm for constrained multi-objective optimization problems with low feasible ratio, *Appl. Soft Comput.* 80 (2019) 42–56.
- [45] S. Zeng, R. Jiao, C. Li, X. Li, J.S. Alkassabeh, A general framework of dynamic constrained multiobjective evolutionary algorithms for constrained optimization, *IEEE Trans. Cybern.* 47 (9) (2017) 2678–2688.
- [46] Y. Zhang, Y. Tian, H. Jiang, X. Zhang, Y. Jin, Design and analysis of helper-problem-assisted evolutionary algorithm for constrained multiobjective optimization, *Inform. Sci.* 648 (2023) 119547.
- [47] A. Zhou, B.-Y. Qu, H. Li, S.-Z. Zhao, P.N. Suganthan, Q. Zhang, Multiobjective evolutionary algorithms: A survey of the state of the art, *Swarm Evol. Comput.* 1 (1) (2011) 32–49.
- [48] Q. Zhu, Q. Lin, J. Chen, A gene-level hybrid search framework for multiobjective evolutionary optimization, *Neural Comput. Appl.* 30 (3) (2018) 759–773.
- [49] E. Zitzler, L. Thiele, Multiobjective optimization using evolutionary algorithms—A comparative case study, in: *International Conference on Parallel Problem Solving from Nature*, Springer, 1998, pp. 292–301.